SE3082

AWS EC2 Lab - Screenshots and Calculations

Lab 08

Exercise 4: Time and speedup graphs

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On instance with 2 vCPUs (t3.micro)

Timing runs - parallel sum code

```
[ec2-user@ip-172-31-17-201 ~]$ ./run_timing.sh sum_mpi sum_times.csv
np=1 avg time=.024753800000000000
np=2 avg time=.013910800000000000
np=4 avg time=.017677000000000000
np=8 avg time=.008429400000000000
```

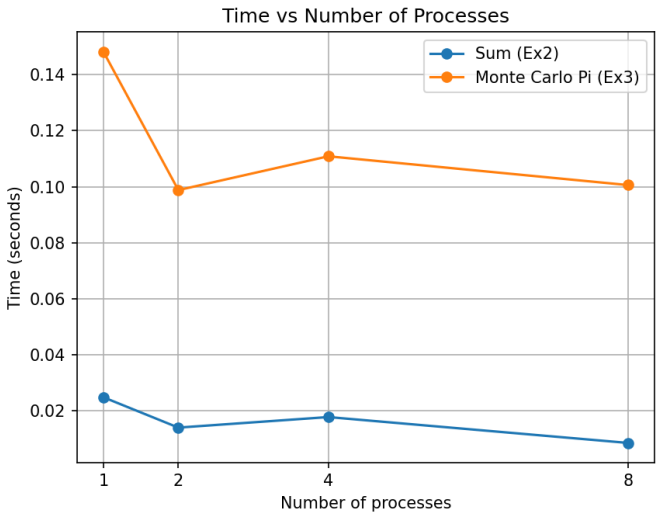
Timing runs - Monte Carlo Pi code

```
[ec2-user@ip-172-31-17-201 ~]$ ./run_timing.sh pi_mpi pi_times.csv
np=1 avg time=.148242000000000000
np=2 avg time=.098770000000000000
np=4 avg time=.110851800000000000
np=8 avg time=.100563000000000000
```

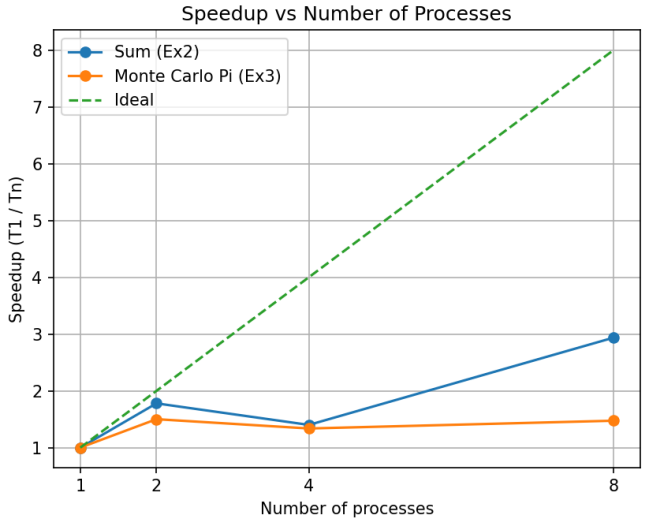
Timing tables - sum and Pi

```
[ec2-user@ip-172-31-17-201 ~]$ cat sum_times.csv pi_times.csv
np,time
1,0.024754
2,0.013911
4,0.017677
8,0.008429
np,time
1,0.148242
2,0.098770
4,0.110852
8,0.100563
```

Time vs number of processes



Speedup vs number of processes



Time and speedup (speedup = T1 / Tn)

Processes	Sum time	Pi time	Sum speedup	Pi speedup
1	0.024754s	0.148242s	1.00	1.00
2	0.013911s	0.098770s	1.78	1.50
4	0.017677s	0.110852s	1.40	1.34
8	0.008429s	0.100563s	2.94	1.47